GOOGLE CLOUD AI BUILDER CUP 2026 · JAPAC

Kairos AI

An expert reliability engineer inside every power device.

Autonomous, remote fault diagnosis and self-healing for critical power infrastructure — catching the moment to act, before downtime.

Theme · Manufacturing

Gemini on Vertex AI

Deployed on Cloud Run

Live: kairos-ai-498775210896.asia-south1.run.app · github.com/puneetpahuja-eaton/kairos-ai

THE ARENA

Build in JAPAC. Build for the world.

The Google Cloud AI Builder Cup invites JAPAC's 15M+ developers to start with a real problem, build a working solution on Google Cloud, and prove that something built for a local need can scale globally.

Our theme

Manufacturing — Intelligent Operations & Industrial Efficiency.

The mandate

Must use Google AI (Gemini / Gemma) and deploy on Cloud Run / GCP / Firebase.

Our answer

Kairos AI — a working, deployed, agentic reliability platform on Google Cloud.

$30,000 prize pool · **In-person Grand Finale in Singapore** · **Judged on Technical Merit & Gen AI (40%), Impact (25%), Innovation (25%), UX (10%)**

THE PROBLEM

When power fails, someone has to be there

In the field, finding a fault still needs a human physically present. By the time an alarm is noticed, reported, and an engineer travels to site, the downtime has already compounded — and expert reliability engineers are scarce and cannot be everywhere at once.

1. No root-cause intelligence

Tools only alert. They don't explain why sags, faults and alarms happen across panelboards and UPS systems — diagnosis stays manual and slow.

2. Manual commissioning & ops

Configuring devices and troubleshooting still need on-site specialists and hand analysis of telemetry — slowing deployments and raising cost.

3. No prediction or correlation

Monitoring lacks early warning and can't correlate cascading alarms along the chain (utility → transformer → UPS → PDU → load), masking the true cause.

4. No closed-loop remediation

Today's systems automate only ~10% of the incident lifecycle (alerting). The other 90% — decide, act, verify — is manual, with no safe agentic workflow.

THE SOLUTION

Kairos AI — the critical moment to act, before downtime

Kairos AI puts an expert reliability engineer inside every device — learning it from its own documentation, watching it remotely 24/7, and closing the loop from detection to verified fix.

Learn

Ingests each device's datasheets, register maps and event-log structure (RAG on Vertex AI).

Monitor

Watches live telemetry and event logs remotely, around the clock.

Diagnose

Explains the root cause in plain English, with cited evidence from the device's own docs.

Act & Verify

Auto-resolves reversible faults or escalates — human-in-the-loop — then verifies the fix.

Seconds, not site visits — the same five-agent engine runs on both live streams and historical data.

WHAT IT DOES

One platform: onboard → learn → analyze → act

Multi-device onboarding

Register any number of devices with role-based access (admin / engineer / viewer).

Learn from anything

Upload PDFs, spreadsheets, notes — or pull from GitHub & Confluence. Embedded with Vertex AI into a per-device knowledge base.

Ask the device

An agentic assistant with Ask / Plan / Agent modes — clarifies first, then generates datasets, test cases, code and monitoring plans.

Analyze, live or historic

A five-agent engine scores uploaded datasets and live REST streams through one code path.

Datasheet-driven live monitoring

The agent reads the datasheet to choose which parameters to poll and decodes event-log codes automatically.

Auto-resolve or escalate

Reversible faults self-heal; the rest escalate to a human — with downtime and cost avoided quantified.

WHERE IT APPLIES

From one panel to an entire fleet

Any device that exposes telemetry or logs — Kairos AI learns it from its own documentation.

DEVICES

- Panelboards
- Power & energy meters
- Switchgear
- UPS & batteries
- PDUs
- Transformers
- Motors & drives
- Breakers

SECTORS WE PROTECT

- Hospitals & clinics
- Water & utilities
- Telecom networks
- Manufacturing plants
- Data centers
- Substations & grid

HIGH-LEVEL ARCHITECTURE

Built end-to-end on Google Cloud



Runs as a dedicated service account via ADC — no API keys embedded in the container. Scales to zero.

THE AGENTS

A closed loop of five agents, guided by a Gemini knowledge plane

Gemini Knowledge & Agentic Plane

RAG retrieval over device docs · Ask / Plan / Agent assistant · dataset & live-monitor-plan generators

1 · Monitor

Detects anomalies in live & historic telemetry (robust-Z + Isolation Forest).

2 · Diagnose

Root cause via topology correlation + Gemini hypotheses with cited evidence.

3 · Decide

Confidence-gated autonomy — weighs reversibility and asset criticality.

4 · Act

Self-heals reversible faults or hands off to a human for approval.

5 · Verify

Confirms the fix worked; scores downtime and cost avoided.

Human-in-the-loop approval gate at Decide → Act

↺ Continuous closed loop — Verify feeds Monitor, so the engine keeps learning

QUANTIFIED IMPACT

Hours of downtime → minutes

Case	Fault (asset)	Path	Downtime avoided	$ avoided
C001	Voltage sag (Main Incomer)	Auto-resolved	40 min	$34,000
C002	Overload (Feeder PDU-2)	Auto-resolved	25 min	$11,250
C003	Battery degradation (UPS-3)	Escalated (early detection)	48 min	$40,800
C004	Thermal runaway (UPS-1)	Auto-resolved	55 min	$46,750
C005	Harmonic distortion (PDU-1)	Auto-resolved	35 min	$15,750
C006	Breaker trip (BRK-4)	Auto-resolved	20 min	$5,000
Total · 6 incidents	**—**	**5 auto / 1 escalated**	**223 min**	**$153,550**

MTTR

tens of hours → 1–2 cycles

Truck-rolls

fewer repeat visits — right-part-first RCA

Monitoring

becomes an outcome-based SLA you can price

Illustrative synthetic scenarios — not realized production savings. Downtime cost rates: critical $850/min, high $450/min, medium $250/min. Every incident is scored against a 'no-platform' baseline.

WHY IT MATTERS

Keeping the lights on for the communities that depend on them

Hospitals, clean-water pumping, vaccine cold-chains, telecom and schools all run on reliable power. Cutting fault response from hours to seconds protects the most vulnerable — especially in remote regions and during climate-stressed grid events.

Essential services uptime

Less downtime at hospitals, water and telecom keeps life-critical services running.

Expertise for everyone

A junior technician — or no one on site — gets senior-engineer diagnosis grounded in the device's own manuals.

Safety

Overload, thermal runaway and harmonic faults are fire and injury hazards; early remote detection prevents accidents.

Sustainability

Catches wasted energy (power factor, harmonics, imbalance), extends equipment life, and cuts diesel-backup and truck-roll emissions.

Resilience

Faster autonomous response hardens critical infrastructure against outages as grids come under stress.

WHY KAIROS AI

Meaningful Gen AI, honest by design

Deep, real Google Cloud AI

Gemini on Vertex AI, Vertex embeddings, Firestore, Cloud Storage, BigQuery, Secret Manager, Cloud Run — not a token API call.

One engine, live + historic

The same five-agent brain diagnoses a replayed dataset and a live warm-up-then-stream — a genuine differentiator.

The agent decides what to watch

It reads the datasheet to pick parameters and thresholds and decodes event logs — monitoring configures itself.

Honest by design

Offline fallbacks are labeled and never shown as Google AI; demo data is synthetic and disclosed; humans approve every action.

121 automated tests · deployed & verified on Cloud Run · Firestore-backed · Gemini on Vertex AI

THE VISION

Start with one device.
Scale to the world.

Kairos AI turns every power device into its own reliability engineer — so the communities that depend on reliable power are never left waiting in the dark.

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